

Regime map: optimiser outcomes by heterogeneity regime

Qualitative synthesis of the tested regimes — not new results, and not a universal optimiser ranking.

Regime	FedAvg	FedProx	FedNova	Key mechanism
IID / near-IID	✓ competitive	~ no gain	✓ competitive	no heterogeneity to exploit
Statistical heterogeneity	✓ competitive	~ fragile gain	— not tested	small per-class rebalancing
L4 straggler (rare-client signal)	✗ collapse	✓ rescue	— not tested	γ -inexact admits the dropped rare-client update
Deterministic LR asymmetry	✗ collapse	✗ collapse	✓ robust	τ -normalisation rescales the weakened client
Random- τ stragglers	✓ competitive	✓ competitive	✗ collapse	τ -normalisation amplifies update noise
Loss-side correction (L4)	✓ competitive	~ no extra gain	— not tested	weighted-CE re-weights the rare classes

✓

 preserved / competitive / rescue / robust

~

 fragile / no extra gain

✗

 collapse / brittle

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 not tested in this thesis